Geometric Group Theory

Problem Sheet 1 - Solutions

Section B

1. Show that the free group of rank r, F_r, has $2^r - 1$ subgroups of index 2. (*hint: consider homomorphisms to* $\mathbb{Z}_2$).

Solution.

Since subgroups of index 2 are normal there is a 1-1 correspondence between such subgroups and onto homomorphisms $f : G \to \mathbb{Z}_2$. But there are $2^r - 1$ onto homomorphisms from F_r to $\mathbb{Z}_2$.

2. i. Show that F_2 has a free subgroup of rank 3.

ii. Show that F_2 has an infinite index free subgroup of rank 2.

iii. Show that F_2 has a free subgroup of infinite rank.

Solution.

i) Let a, b be a basis of F_2. $x_1 = aba^{-1}, x_2 = a^2ba^{-2}, x_3 = a^3ba^{-3}$ generate a free subgroup of F_2. Indeed one sees easily by induction on the length that reduced words on x_1, x_2, x_3 correspond to non trivial reduced words on a, b. Precisely we claim that a reduced word v of length k on x_1, x_2, x_3 gives a reduced word w on a, b with k occurrences of $b^{\pm 1}$. Further if the last letter of v is $x_i^{\pm 1}$ the last letters of w are $b^{\pm 1}a^{-i}$. One verifies the inductive step case by case.

It follows that the obvious homomorphism $\varphi : F(x, y, z) \to F(a, b)$ is 1-1, ie $< x, y, z >$ is a free subgroup of rank 3.

ii) $< x, y >$ is an infinite index subgroup of $< x, y, z >$ hence of F_2.

iii) The set a^nba^{-n} $n \in \mathbb{N}, n > 0$ is a free basis of an infinitely generated free subgroup of F_2 as in i).

3. Prove the ping-pong lemma: Let G be a group acting on a set S and let $a, b \in G$ be infinite order elements. If there are non empty disjoint subsets A, B of S such that $a^nB \subseteq A$, $b^nA \subseteq B$ for all $n \in \mathbb{Z} \setminus \{0\}$ then $\{a, b\}$ generate a free subgroup of G. (*hint: if* $w = a^{k_1}b^{k_2}...a^{k_n}$ *then show that* $wB \subseteq A$. *Otherwise replace* w *by a conjugate and use the same argument*).
Solution.

Clearly if $w = a^{k_1}b^{k_2}...a^{k_n}$ with $k_1, k_n \neq 0$ then $wB \subseteq A$ so $w \neq 1$. Now if a reduced word w in a, b is not of this form then a^nwa^{-n} is of this form for some n. It follows that $a^nwa^{-n} \neq 1$ hence $w \neq 1$. It follows that the obvious homomorphism $\varphi : F(a, b) \to < a, b >$ is 1-1 so $< a, b >$ is a free group of rank 2.

4. Show that the matrices

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

generate a free subgroup of $SL_2(\mathbb{Z})$. (*hint: apply the ping pong lemma to the usual action of* $SL_2(\mathbb{Z})$ *on* $\mathbb{R}^2$ *with* $A = \{(x, y) : |x| > |y|\}$, $B = \{(x, y) : |x| < |y|\}$). *Solution.* We set

$$a = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \ b = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

and we note that

$$a^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}, \ b^n = \begin{pmatrix} 1 & 0 \\ 2n & 1 \end{pmatrix}$$

Clearly $a^n B \subseteq A$, $b^n A \subseteq B$ and A, B disjoint, so we just apply the ping pong lemma.

5. i) Let $G_1 = \langle S_1 | R_1 \rangle$, $G_2 = \langle S_2 | R_2 \rangle$. Find a presentation for the direct product $G_1 \times G_2$.

ii) If $G = \langle S | R \rangle$ find a presentation for the abelianization of G.

Solution. i) Let $H = \langle S_1 \cup S_2 | R_1 \cup R_2 \cup \{st = ts : s \in S_1, t \in S_2\} \rangle$. Then define a map $f : H \to G_1 \times G_2$ by $s \to (s, e), t \to (e, t)$, $s \in S_1, t \in S_2$. This extends to a homomorphism since relators map to 1. It is clearly onto. Finally if $f(w) = 1$ then $w = vu$ where v is a word on S_1 and u a word on S_2 so $v = 1, u = 1$ and $w = 1$.

ii) Let $H = \langle S | R \cup \{st = ts : s, t \in S\} \rangle$. The obvious maps extending $s \to s, s \in S$ from $H \to G/[G, G]$, $\quad G/[G, G] \to H$ are inverse to each other.

6. Show that the group $G = \langle a, b | ababa = 1 \rangle$ is abelian.

Solution. $(ab)^2 = a^{-1}$, $(ba)^2 = a^{-1}$ and $(ab)^3 = b^{-1}$, $(ba)^3 = b^{-1}$ so $(ab)^2 = (ba)^2$, $(ab)^3 = (ba)^3$. It follows that $ab = ba$.

7. i. Show that the group

$$G =< x, y | x^2 = y^3 >$$

is not trivial.

ii. Show that the group G is isomorphic to the group $H =< a, b | aba = bab >$

Solution.

i) Clearly we can map onto $\mathbb{Z}$: $x \to 3$, $y \to 2$.

ii)

$$\langle a, b | aba = bab \rangle \to \ (T2) \ \langle a, b, x, y | y = ab, x = aba, aba = bab \rangle$$

But now we can eliminate a, b using $T2$ again since $a = yb^{-1}$ and $x = bab = by \implies b = xy^{-1}$ and we get the presentation $\langle x, y | x^2 = y^3 \rangle$.

Alternatively we may define $\phi : G \to H$ by $\phi(x) = aba, \phi(y) = ab$ and $\psi : H \to G$ by $\psi(a) = y^{-1}x, \psi(b) = xy^{-1}$ and check that $\phi \circ \psi(a) = a, \phi \circ \psi(b) = b$ and $\psi \circ \phi(x) = x, \psi \circ \phi(y) = y$ so ϕ is an isomorphism.

8. Show that every finitely presented group has a presentation in which every relation is a word of length at most 3.

Solution. If r is a relation of length ≥ 4 then $r = r_1 r_2$ (reduced word) with $|r_1| = 2, |r_2| \geq 2$. Using Tietze transformation introduce a new generator a_1 and the realtor $r_1 a_1^{-1}$. Now replace r by ar_2. We repeat this till all relators have length ≤ 3.

9. If H is a finitely generated subgroup of G then the *membership problem* for H asks whether there is an algorithm to decide if $g \in G$ lies in H. Show that the membership problem is solvable for cyclic subgroups of F_n (the free group of rank n). In other words there is an algorithm such that given $u, w \in F_n$ decides whether $u \in < w >$.

Solution. If $w = ava^{-1}$ with v cyclically reduced it is enough to check whether $u = w^n$ for all $n \leq |u|$.

Section C

10. An infinite finitely generated group is called almost finite if all its quotients are finite groups. Show that every infinite finitely generated group has a quotient that is almost finite.

Solution Let $G =< S|R >$ be a given f.g. infinite group. We enumerate all words on S and we go through the list asking whether adding w_i to relations R results to a finite group. If it does not we add it, if not we go to the next word. If this stops we get the quotient we need. If it goes on forever we add all these countably many relators. We remark that the group we obtain is infinite. This is because finite groups are finitely presented so if it were finite we would have already all the relators in a finite stage. But now we can not add more relators so the quotient we have has the required property.

Alternatively order normal subgroups N such that G/N is infinite by inclusion. An ascending union of such subgroups has the same property since if G/N is finite it is finitely presented so we can find all its relations in a fixed subgroup in the union. So by Zorn's lemma G/N is infinite. On the other hand any quotient of it is finite.

11. Show that the following presentations are presentations of the trivial group:

i) $\langle a, b, c | aba^{-1} = b^2, bcb^{-1} = c^2, cac^{-1} = a^2 \rangle$

ii) $\langle a, b | a^n = b^{n+1}, aba = bab \rangle$
iii) $\langle a, b | ab^n a^{-1} = b^{n+1}, ba^n b^{-1} = a^{n+1} \rangle$.

Solution.

I am fairly certain the solutions below are not the shortest possible.

i) We note that we have the relations:

$$a^k b a^{-k} = b^{2^k},\ b^{-1}ab = ba,\ c^{-1}bc = cb,\ a^{-1}ca = ac$$

We have

$$cac^{-1}bca^{-1}c^{-1} = b^4,\ b^4cb^{-4} = c^{16}$$

so

$$(cac^{-1}bca^{-1}c^{-1})c(cac^{-1}bca^{-1}c^{-1})^{-1} = c^{16} \Rightarrow$$

$$a(c^{-1}b)ca^{-1}(cac^{-1})b^{-1}ca^{-1} = c^{16} \Rightarrow ac(bab^{-1})ca^{-1} = c^{16} \Rightarrow$$

$$ac(bab^{-1})ca^{-1} = c^{16} \Rightarrow acb^{-1}(aca^{-1}) = c^{16} \Rightarrow$$

$$acb^{-1}a^{-1}c = c^{16} \Rightarrow acb^{-1}a^{-1} = c^{15} \Rightarrow (aca^{-1})(ab^{-1}a^{-1}) = c^{15} \Rightarrow$$

$$a^{-1}cb^{-2} = c^{15} \Rightarrow a^{-1}b^{-2}(b^2cb^{-2}) = c^{15} \Rightarrow a^{-1}b^{-2} = c^{11} \Rightarrow b^2a = c^{-11} \Rightarrow ab = c^{-11}$$

Now

$$c^{11}ac^{-11} = a^{2^{11}} \Rightarrow b^{-1}a^{-1}aab = a^{2^{11}} \Rightarrow b^{-1}ab = a^{2^{11}} \Rightarrow ba = a^{2^{11}} \Rightarrow b = a^{2^{11}-1}$$

But then $aba^{-1} = b^2 \Rightarrow b = 1 \Rightarrow c = 1 \Rightarrow a = 1$.

ii) Using $aba = bab$ and induction we get $a^nba = bab^n$.

$$b^{n+1}ab = b^nbab = b^naba = ... = aba^{n+1}$$

$$aba^{n+1} = aba^na = ab^{n+2}a = a^{n+1}ba$$

So

$$b^{n+1}ab = a^{n+1}b = a^{n+1}ba \Rightarrow a = 1 \Rightarrow b^2 = b \Rightarrow b = 1$$

iii) We remark that $ba^{nk}b^{-1} = (ba^nb^{-1})^k = a^{k(n+1)}$. So $b^na^{n^n}b^{-n} = a^{(n+1)^n}$. Now

$$(ab^na^{-1})a^{n^n}(ab^na^{-1})^{-1} = ab^na^{n^n}b^{-n}a^{-1} = a^{(n+1)^n}$$

and

$$(ab^na^{-1})a^{n^n}(ab^na^{-1})^{-1} = b^{n+1}a^{n^n}b^{-(n+1)} = ba^{(n+1)^n}b^{-1}$$

Therefore

$$ba^{(n+1)^n}b^{-1} = a^{(n+1)^n} \Rightarrow ba^{n(n+1)^n}b^{-1} = a^{n(n+1)^n} \Rightarrow a^{(n+1)^{n+1}} = a^{n(n+1)^n}$$

It follows that

$$a^{(n+1)^n} = 1$$

$$ba^n b^{-1} = a^{n+1} \Rightarrow (ba^n b^{-1})^{(n+1)^{n-1}} = a^{(n+1)^n} \Rightarrow a^{n(n+1)^{n-1}} = 1$$

It follows that

$$a^{n(n+1)^{n-1}} = a^{(n+1)^n} \Rightarrow a^{(n+1)^{n-1}} = 1$$

We continue inductively and we get

$$a^{n+1} = 1 \Rightarrow a^n = 1 \Rightarrow a = 1 \Rightarrow b = 1$$